On the other hand, viewing $\mathcal{L}_f$ as an operator on the phase variables only, we have

$$\begin{aligned}(\mathcal{L}_f \tilde{g})(z) &= \boldsymbol{f}(z) \cdot \nabla_z \left[g(\mathscr{F}_t(z))\right] \\ &= \boldsymbol{f}(z) \cdot \left[\mathscr{F}_t'(z)^T \nabla g(\mathscr{F}_t(z))\right] \\ &= \left[\mathscr{F}_t'(z)\boldsymbol{f}(z)\right] \cdot \nabla g(\mathscr{F}_t(z)) \\ &= \boldsymbol{f}(\mathscr{F}_t(z)) \cdot \nabla g(\mathscr{F}_t(z)) \\ &= (\mathcal{L}_f g)(\mathscr{F}_t(z)).\end{aligned}$$

This is the same as the right hand side of (5.1). This means that $\tilde{g}$ satisfies the equation

$$\frac{\partial \tilde{g}}{\partial t} = \mathcal{L}_f \tilde{g}. \tag{5.2}$$

This is a time-dependent, linear (hyperbolic) partial differential equation. The steady state solutions of this equation (defined by $\partial \tilde{g}/\partial t = 0$) are of the form $\tilde{g}(z,t) = \varphi(z)$ where φ satisfies $\mathcal{L}_f \varphi(z) \equiv 0$. These are just the first integrals of the ordinary differential equation system $\dot{z} = \boldsymbol{f}(z)$.

It is intuitively appealing to define the solution of (5.2) by

$$\tilde{g}(z,t) = G_t \tilde{g}(z,0), \qquad \text{where} \quad G_t = e^{t\mathcal{L}_f}.$$

Let us refer to G_t as the *solution operator* associated to the differential equation $\dot{z} = \boldsymbol{f}(z)$. Calculating the exponential operator $e^{t\mathcal{L}_f}$ can be thought of as tantamount to solving a linear partial differential equation for given (Cauchy) initial data.

The projection onto the ith coordinate, denoted ∂_i, is itself a scalar valued function of z ($\partial_i(z) = z_i$). Calculating directly we have $\mathcal{L}_f z_i = f_i$. The interpretation of the solution of the initial value problem in terms of the solution operator G_t is given by

$$\left(e^{t\mathcal{L}_f} \partial_i\right)(z(0)) = z_i(t).$$

The notion of the solution operator and the concept of flow map are essentially equivalent descriptions of the dynamics of the system.

In physics treatments, the Liouvillian is often denoted $i\mathcal{L}$. This may seem natural since (a) it allows a formal correspondence with the Schrödinger equation and between the propagators of quantum mechanics and classical mechanics, and (b) for Hamiltonian dynamics, as we shall see, the Liouvillian is skew-adjoint (in a certain sense skew symmetric) and the inclusion of i explicitly calls attention to this fact. However, we feel it is more natural to omit the i in a treatment that includes study of both stochastic and deterministic models.